C01 AT2 CONCEPT MAPPING

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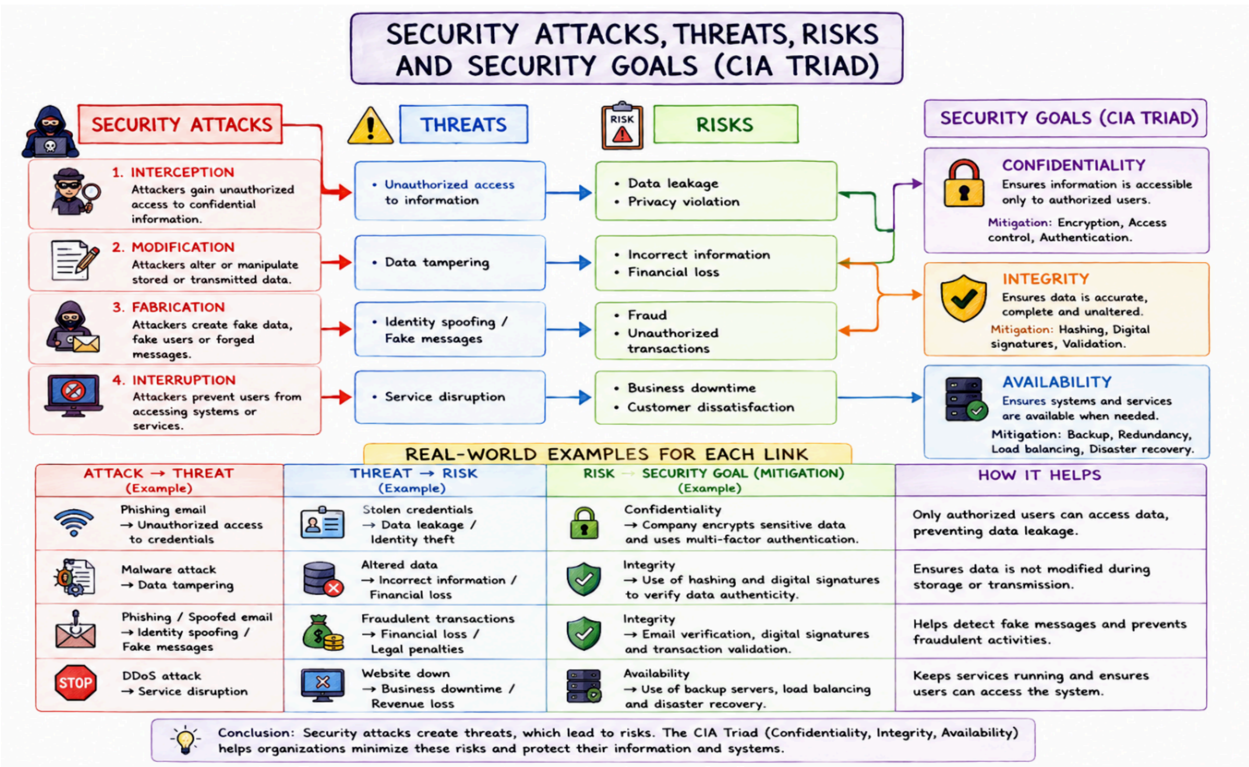
1. **Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:**

   a. Show how each attack leads to specific threats and risks

   b. Map how security goals help mitigate them Provide one real-world example for each link

Answer :

Concept Map – Security Attacks, Threats, Risks and Security Goals



(a) How Each Attack Leads to Threats and Risks

Security Attack	Threat	Risk
Phishing Attack	Unauthorized access to user credentials	Data theft, identity theft, financial loss
Malware Attack	Corrupts files and compromises systems	Data loss, system failure, financial damage
Denial of Service (DoS)	Prevents users from accessing services	Service downtime and business interruption

(b) Mapping Security Goals to Mitigate Threats and Risks

Security Goal	Mitigates	How it Protects
Confidentiality	Unauthorized access and data theft	Uses encryption, authentication, and access control to keep information private.
Integrity	Data modification or corruption	Uses hashing, digital signatures, and validation to ensure data remains accurate.
Availability	Service disruption and downtime	Uses backups, redundancy, load balancing, and disaster recovery to keep services available.

(c) Real-World Examples

1. Attack → Threat

- Attack: Phishing email
- Threat: Unauthorized access to employee credentials
- Example: An attacker sends a fake Microsoft login page to steal an employee's password.

2. Threat → Risk

- Threat: Stolen login credentials
- Risk: Confidential customer information is leaked.

- Example: Hackers use stolen banking credentials to access customer accounts and perform fraudulent transactions.

3. Risk → Security Goal (Mitigation)

Risk	Security Goal	Real-World Example
Data leakage	Confidentiality	A hospital encrypts patient records so only authorized doctors can access them.
Data tampering	Integrity	Online banking uses digital signatures to ensure transaction details are not altered.
Website outage	Availability	An e-commerce website uses backup servers and load balancing to remain accessible during a DDoS attack.

Conclusion

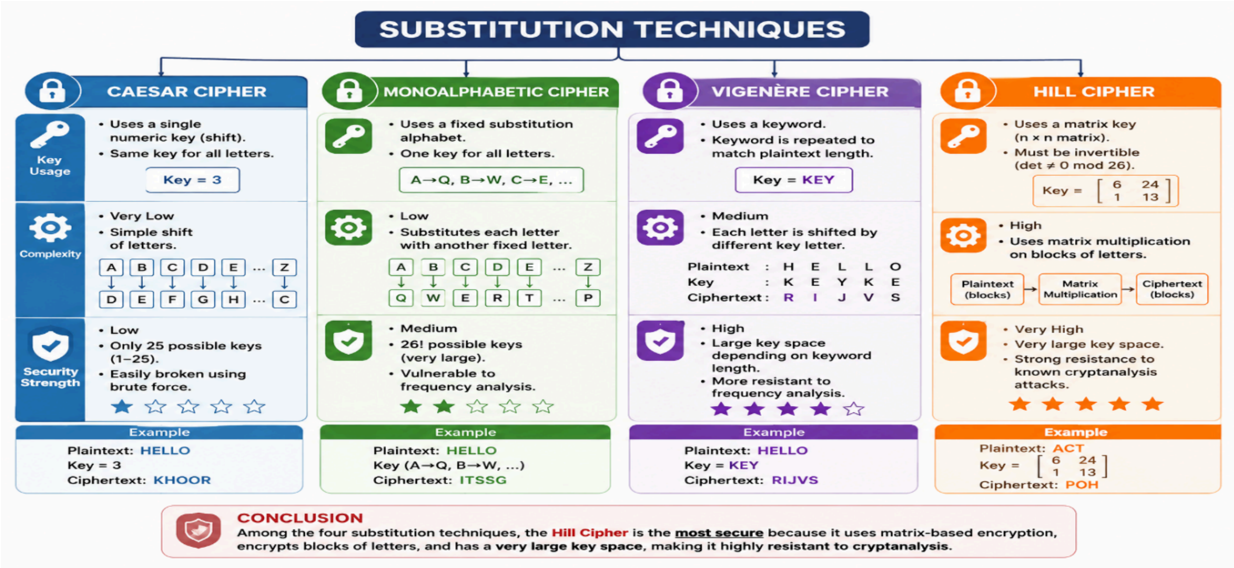
Security attacks create threats, which lead to risks affecting an organization's information and services. The CIA Triad helps reduce these risks:

- Confidentiality protects sensitive information from unauthorized access.
- Integrity ensures that information remains accurate and unaltered.
- Availability ensures that systems and services remain accessible to authorized users.

2. **Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.**

Answer :

Concept Map – Comparison of Substitution Techniques



Comparison Based on Key Usage, Complexity and Security Strength

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Uses a single fixed shift key (e.g., +3)	Very Low	Low
Monoalphabetic Cipher	Uses one fixed substitution alphabet	Low	Medium
Vigenère Cipher	Uses a repeating keyword (e.g., KEY)	Medium	High
Hill Cipher	Uses an invertible matrix as the encryption key	High	Very High

Analysis of Each Technique

1. Caesar Cipher

- Uses one shift value for every letter.
- Easy to encrypt and decrypt.
- Easily broken using brute-force or frequency analysis.

2. Monoalphabetic Cipher

- Each plaintext letter is replaced by another fixed letter.
- More secure than Caesar Cipher.
- Still vulnerable to frequency analysis.

3. Vigenère Cipher

- Uses a repeating keyword.
- Different letters are shifted by different amounts.
- More resistant to frequency analysis than Caesar and Monoalphabetic ciphers.

4. Hill Cipher

- Uses matrix multiplication for encryption.
- Encrypts multiple letters together as blocks.
- Difficult to break without knowing the matrix key.

Most Secure Technique

Hill Cipher

Reason

- Uses matrix-based encryption, making the relationship between plaintext and ciphertext more complex.
- Encrypts blocks of letters instead of individual characters.
- Provides higher resistance to frequency analysis than the other three techniques.
- The matrix key significantly increases the number of possible keys, making brute-force attacks much more difficult.

Conclusion

Among the four substitution techniques:

- Caesar Cipher – Least secure because it uses a single shift.

- Monoalphabetic Cipher – More secure than Caesar but still vulnerable to frequency analysis.
- Vigenère Cipher – Uses a repeating keyword, providing better security than simple substitution methods.
- Hill Cipher – Most secure because it employs matrix-based block encryption, making cryptanalysis much more difficult than the other classical substitution techniques.

3. **Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:**

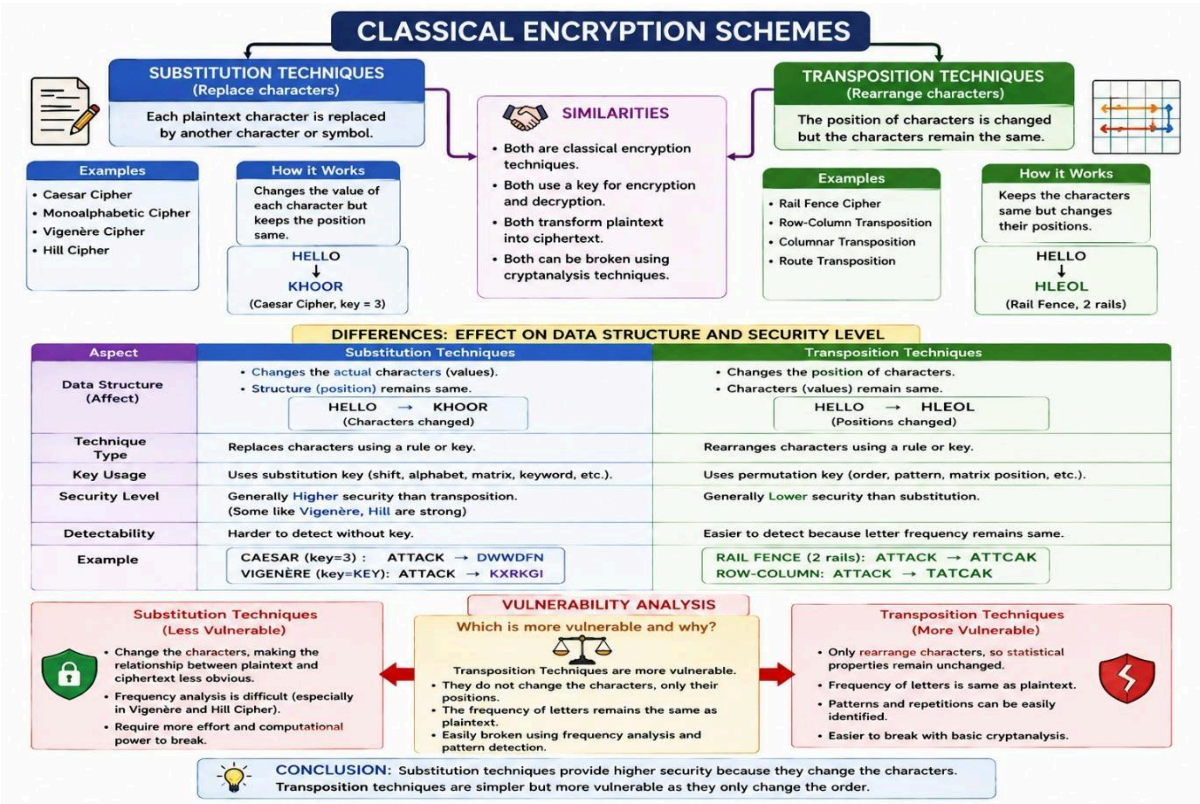
   a. Show differences and similarities

   i.Map how each affects the Data structure and Security level

   ii.Analyze which is more vulnerable to attacks and why

Answer :

Concept Map – Classical Encryption Schemes



(a) Differences and Similarities between Substitution and Transposition Techniques

Similarities

- Both are classical encryption techniques used to protect information.
- Both convert plaintext into ciphertext.
- Both require a key for encryption and decryption.
- Both aim to prevent unauthorized access to data.

Differences

Aspect	**Substitution Techniques**	**Transposition Techniques**
Method	Replaces each plaintext character with another character.	Rearranges the positions of characters without changing them.
Data Structure	Character values change, but their positions remain the same.	Character values remain the same, but their positions change.
Examples	Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher, Hill Cipher	Rail Fence Cipher, Columnar Transposition Cipher
Security Level	Generally higher security.	Comparatively lower security.

(b) Effect on Data Structure and Security Level

Substitution Techniques

- Replace one character with another according to a key.
- The character values are changed.
- The order of characters remains the same.
- Security is stronger because attackers cannot directly identify the original letters.

Example

Plaintext:

HELLO

Caesar Cipher (Shift = 3):

KHOOR

Here, the letters are changed but their positions remain unchanged.

Transposition Techniques

- Rearrange the order of characters.
- The character values remain unchanged.
- Only the positions of the characters are altered.
- Easier to analyze because letter frequencies remain the same.

Example

Plaintext:

HELLO

Rail Fence Cipher (2 Rails):

HLEOL

The letters remain the same but appear in a different order.

(c) Which Technique is More Vulnerable? Why?

More Vulnerable: Transposition Techniques

Reasons

- Only the positions of letters are changed.
- The frequency of letters remains identical to the original plaintext.
- Attackers can use frequency analysis and pattern recognition to recover the message.
- It offers limited protection against modern cryptanalysis.

Less Vulnerable: Substitution Techniques

- Character values are replaced with different symbols or letters.
- Frequency analysis becomes more difficult, especially in Vigenère Cipher and Hill Cipher.
- Provides better confidentiality than transposition methods.

Conclusion

- Substitution Techniques change the characters while keeping their positions the same, resulting in higher security.
- Transposition Techniques only rearrange the positions of characters, making them more vulnerable to cryptanalysis.
- Therefore, Substitution Techniques are generally more secure, whereas Transposition Techniques are easier to break because they preserve the original character frequencies.

4. **Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:**

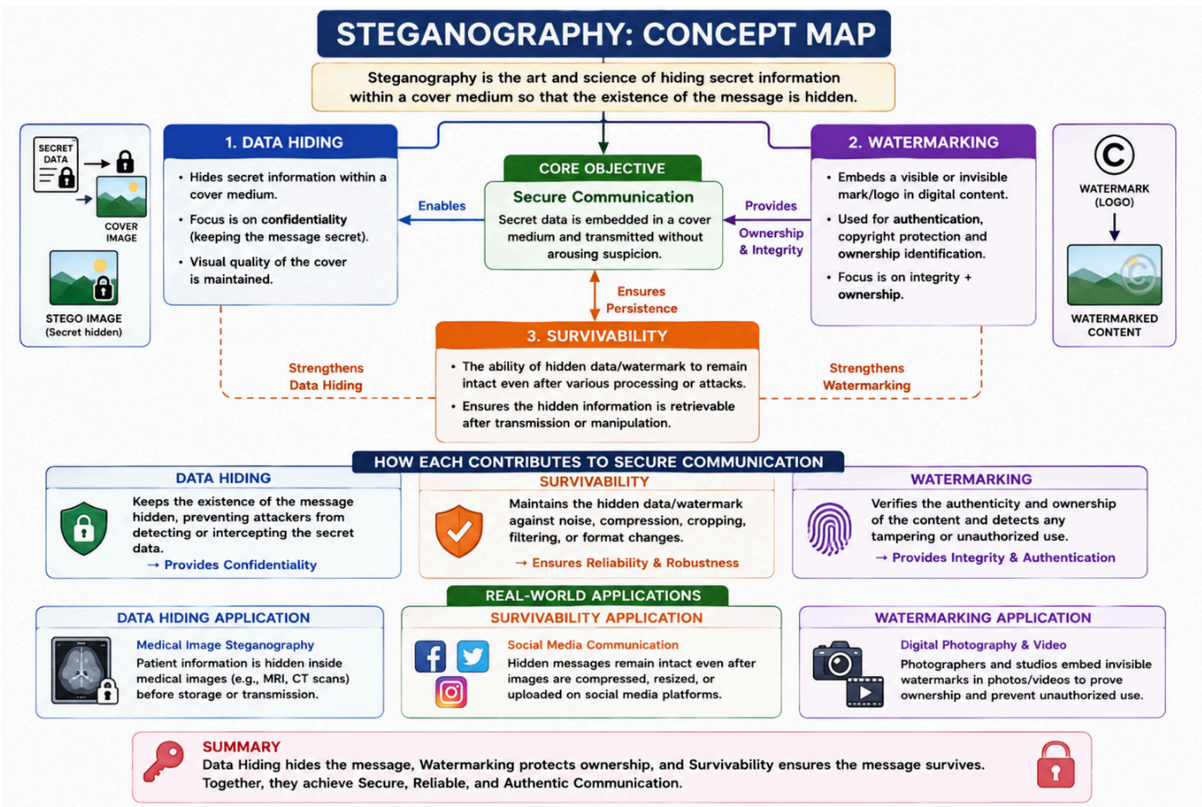
   a. Show relationships between these concepts

   b. Analyze how each contributes to secure communication

   c. Include one real-world application for each

Answer :

Concept Map – Steganography



(a) Relationship between Data Hiding, Watermarking, and Survivability

Steganography

Steganography is the process of hiding secret information inside a cover medium (such as an image, audio, or video) so that the existence of the message is concealed.

The three important concepts related to steganography are:

1. Data Hiding

- Data hiding is the process of embedding confidential information inside a digital file.
- Its main objective is to ensure that the hidden message cannot be noticed by unauthorized users.

2. Watermarking

- Watermarking embeds a visible or invisible mark into digital content.
- It is mainly used to verify ownership, authenticity, and copyright protection.

3. Survivability

- Survivability refers to the ability of the hidden data or watermark to remain intact even after image compression, editing, resizing, or transmission.
- It ensures that the embedded information can still be recovered.

Relationship

- Steganography is the overall technique.
- Data Hiding conceals confidential information.
- Watermarking protects ownership and authenticity.
- Survivability ensures that the hidden information remains available even after modifications.

(b) How Each Contributes to Secure Communication

Concept	**Contribution to Secure Communication**

Data Hiding	Protects confidential information by hiding it inside a cover file, making it difficult for attackers to detect.
Watermarking	Verifies the authenticity and ownership of digital content, preventing unauthorized copying or modification.
Survivability	Ensures that the hidden data or watermark remains intact even after compression, resizing, filtering, or transmission.

(c) Real-World Applications

1. Data Hiding

Application: Medical Image Security

- Hospitals hide patient information inside MRI or CT scan images before storing or transmitting them.
- This protects patient privacy while maintaining the image quality.

2. Watermarking

Application: Copyright Protection of Digital Images

- Photographers and media companies embed invisible watermarks into images and videos.
- This proves ownership and prevents unauthorized copying or distribution.

3. Survivability

Application: Secure Social Media Communication

- Hidden messages remain recoverable even after images are compressed or resized by platforms such as WhatsApp, Facebook, or Instagram.
- This ensures reliable communication despite image processing.

Conclusion

Steganography combines Data Hiding, Watermarking, and Survivability to provide secure communication.

- Data Hiding keeps confidential information secret.

- Watermarking protects ownership and authenticity.
- Survivability ensures that the hidden information remains accessible even after image modifications.

5. **Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:**
    a. Show how each concept is mathematically connected

    b. Map their role in encryption/decryption

    c. Analyze which concept is most critical for public key cryptography and justify

Answer :

Concept Map – Number Theory Concepts Used in Cryptography

(a) Mathematical Connection Between the Concepts

Concept Map Explanation

- Extended Euclidean Algorithm is used to find the Greatest Common Divisor (GCD) of two numbers.
- If the GCD of two numbers is 1, they are coprime.
- When two numbers are coprime, the Extended Euclidean Algorithm can compute the Modular Inverse.
- Euler's Phi Function ($\varphi(n)$) calculates the number of integers less than *n* that are relatively prime to *n*.
- Euler's Theorem uses Euler's Phi Function and states: $a^{\varphi(n)} = 1 \pmod p$
- Fermat's Little Theorem is a special case of Euler's Theorem when n is a prime number.

  $a^{(p-1)} = 1 \pmod p$

Concept Relationship

Extended Euclidean Algorithm

|

▼

Finds Modular Inverse

|

▼

Used in Encryption & Decryption

▲

|

Euler's Phi Function ——► Euler's Theorem

|

▼

Fermat's Little Theorem

(Special case when p is prime)

(b) Role of Each Concept in Encryption and Decryption

Concept	Role in Cryptography
Extended Euclidean Algorithm	Computes the GCD and finds the modular inverse required for RSA private key generation.
Modular Inverse	Used to calculate the private key during RSA decryption.
Euler's Phi Function ($\varphi(n)$)	Determines the number of integers relatively prime to *n* and is used in RSA key generation.

Euler's Theorem	Provides the mathematical foundation that guarantees correct RSA encryption and decryption.
Fermat's Little Theorem	Simplifies modular arithmetic and is used in prime number testing and cryptographic computations.

(c) Most Critical Concept for Public Key Cryptography

Most Critical Concept: Euler's Phi Function ($\varphi(n)$)

Justification

- Euler's Phi Function is essential for generating RSA public and private keys.
- It is used to calculate the private key (d) from the public key (e) using the modular inverse.
- Without $\varphi(n)$, RSA key generation cannot be performed.
- Euler's Theorem depends directly on Euler's Phi Function, making it the foundation of RSA encryption and decryption.

Therefore, Euler's Phi Function is considered the most critical concept in public key cryptography.

Conclusion

The number theory concepts are closely related and together provide the mathematical foundation for modern public key cryptography.

- Extended Euclidean Algorithm → Finds the Modular Inverse.
- Euler's Phi Function → Determines the number of coprime integers.
- Euler's Theorem → Forms the basis of RSA encryption and decryption.
- Fermat's Little Theorem → A special case of Euler's Theorem used in modular arithmetic.
- Modular Inverse → Enables computation of the RSA private key.